

2018-Ph-H1-Q16

Section: Electricity

Topic: Capacitors

Which graph shows the voltage across a discharging capacitor?

Solution:

When a capacitor discharges, the voltage drops **exponentially** — not in a straight line — approaching zero but never quite reaching it.

So the correct graph is the one showing a **curve starting at maximum voltage and falling steeply at first, then flattening** over time.

Answer: The exponential decay graph (Option C)

Revision Tip:

Capacitor discharging:

- Voltage and current both decrease exponentially
- Use $V(t) = V_0 e^{-t/RC}$ if formulas are needed in later questions